



SYDNEY BOYS HIGH SCHOOL

2020

YEAR 9
TERM 4
YEARLY ASSESSMENT TASK

Mathematics Stage 5.3

General Instructions

- All questions may be attempted.
- Write using black or blue pen.
- Marks may be deducted for careless or badly arranged work. This can include insufficient working out.
- All working and answers are to be written in this test booklet.
- NESA approved calculators may be used.
- Clearly indicate your class by placing an **X** next to your class.
- If you wish to rewrite an answer, draw a line through your faulty answer and rewrite your answer on Extra working pages provided.
- Show the number and part of the answer being rewritten and write “**See Back**” at the original question.

Time Allowed: 90 minutes

Examiner: JM

Name: _____

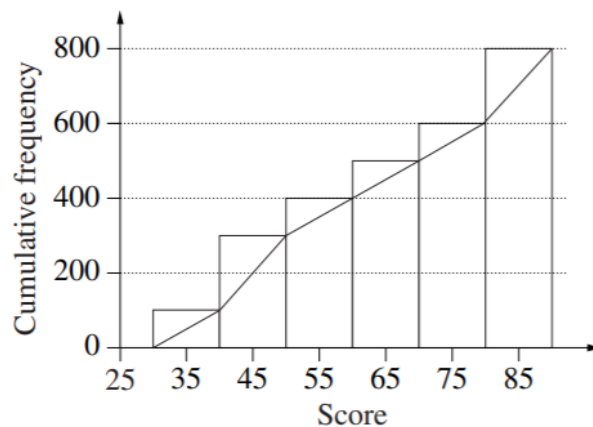
Class	Teacher	
9MaA	Ms B Kilmore	
9MaB	Mr T Brockman/Ms A Ward	
9MaC	Ms Z Zeng	
9MaP	Mr S Gurjar/ Mr A Wang	
9MaL	Mr R Wang	
9MaU	Mr AM Gainford/ Mr D Taylor	
9MaS	Mr J James	

Parts	Marks
A	/22
B	/22
C	/24
D	/24
E	/22
Total	/114

Part A (22 marks)

For Questions 1 – 6, Circle the correct answer.

1. Stephen the author earns a percentage of the sales of his recent horror novel. Which word best describes this payment? 1
- (A) Salary
- (B) Commission
- (C) Royalty
- (D) Wage
2. The gradient and the y -intercept, respectively, of the line $2x + y - 3 = 0$ are? 1
- (A) 3 and -2
- (B) -2 and 3
- (C) 2 and -3
- (D) -3 and -2
3. A set of data is represented by the cumulative frequency histogram and ogive. 1



What is the best approximation for the interquartile range for this set of data?

- (A) 35
- (B) 30
- (C) 25
- (D) 40

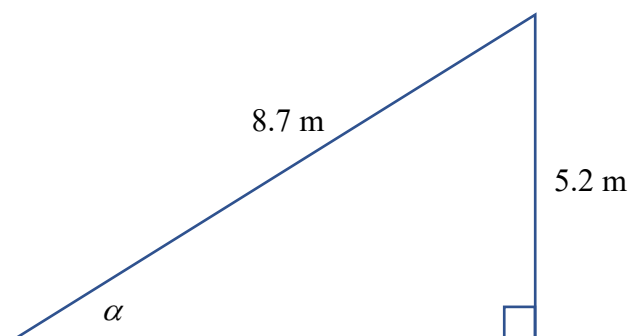
4. Carlos took 70% of a whole pie. Maria took one third of the remainder. What percentage of the whole pie is left?

1

- (A) 30%
(B) 15%
(C) 10%
(D) 20%

5. Find the size of the angle α correct to the nearest minute.

1



- (A) $30^{\circ}52'$
(B) $30^{\circ}53'$
(C) $36^{\circ}42'$
(D) $36^{\circ}43'$

6. Simplify $2m^2 \times 3mp^2$.

1

- (A) $5m^2p^2$
(B) $5m^3p^2$
(C) $6m^2p^2$
(D) $6m^3p^2$

7. Express $9\,876\,000 \times 1\,234\,000$ in scientific notation correct to 3 significant figures. 2

8. Simplify $9hk - j^2 + 5kh + 4j^2$. 1

9. Expand and simplify $x - 3y - 4(y - 2x)$. 1

10. Calculate $\frac{(3.2)^3}{6.8 + 4.7}$ correct to 4 decimal places. 1

11. A farmer spreads 25 kg of fertiliser over an area of 4000 m^2 . How much fertiliser would be needed to cover an area of 1.5 ha? 1

12. A car uses 18 L of petrol to travel 150 km. How much petrol would be needed to travel 350 km? 1

13. Given that $a * b$ denotes $a^2 + ab - b^2$, find the value of $5 * (-2)$. 1

14. Simplify

(a) $\frac{3x}{5} - \frac{x}{5}$ 1

(b) $\frac{8a - 4b}{4}$ 1

(c) $\frac{2x - 3}{5} - \frac{x - 1}{2}$ 2

(d) $\frac{1}{x} + \frac{2}{x+1}$

2

(e) $\frac{1}{2x+1} - \frac{1}{5x-1}$

2

End of Part A

Part B (22 marks)

1. Solve $5 - 2x > 13$, and display your solution on a number line.

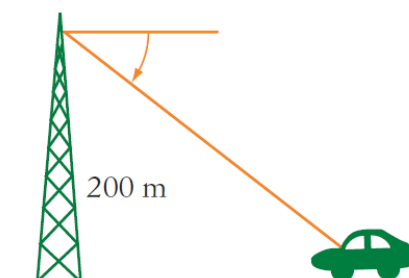
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2. If $16m^2 + 24m + P = (4m + Q)^2$, what is the value for P ?

2

3. From the top of a 200 metre tower, the angle of depression of a car is 49° . How far (to the nearest metre) is the car from the base of the tower?

1

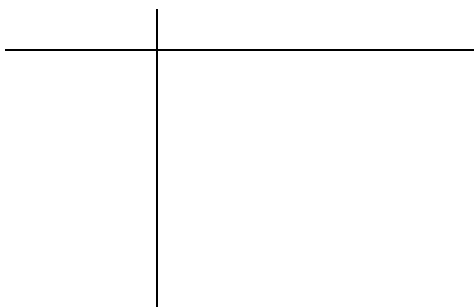


4. At a tourist information office, the numbers of people seeking information each hour over the course of a 12-hour day are shown below.

6 25 38 39 31 18 35 31 33 15 21 28

- (a) Construct a sorted stem and leaf diagram to represent these data.

1



- (b) State the type of skewness suggested by your stem and leaf diagram.

1

- (c) For these data find the median, the mean and the mode.
Comment on the usefulness of the mode in this case.

4

5. Simplify and write without negative indices.

(a) $\left[-\left(\frac{1}{a}\right)^{-2} \right]^{-3}$ 2

(b) $\frac{(27a^2b^{-3})^{-1}}{(15ab^{-2})^{-2}}$ 2

6. Expand and simplify

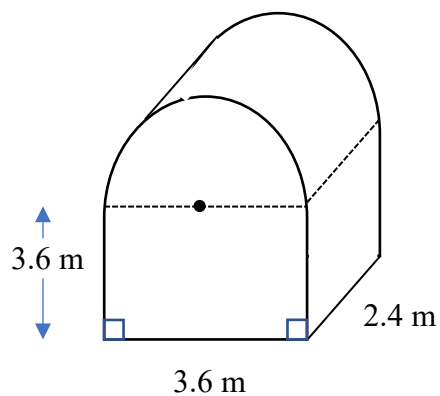
(a) $(x+5)(2-3x)$ 1

(b) $(2y+3)^2$ 1

(c) $(2x-3)(2x+3)$ 1

7. Calculate the surface area and volume of this enclosed solid, correct to 1 decimal place.

4



End of Part B

Part C (24 marks)

1. Solve $y = 4x + 3$ and $y = 17 - 3x$ simultaneously.

2

2. If $y - \frac{1}{y} = 2\sqrt{6}$, then evaluate $y^2 + \frac{1}{y^2}$.

2

3. If $y = 5^x$. Express 5^{2x+1} in terms of y .

2

4. The table shows the income tax rates for the 2019–2020 financial year.

3

Taxable income	Tax payable on this income
\$0 to \$18 200	Nil
\$18 201 to \$37 000	19c for each \$1 over \$18 200
\$37 001 to \$90 000	\$3 572 plus 32.5c for each \$1 over \$37 000
\$90 001 to \$180 000	\$20 797 plus 37c for each \$1 over \$90 000
\$180 001 and over	\$54 097 plus 45c for each \$1 over \$180 000

(Note: The above rates do not include the Medicare levy of 2%.)

The Medicare levy is calculated as 2% of taxable income.

For the 2019–2020 financial year, Charlotte pays a Medicare levy of \$2341.20.

Calculate the tax payable on Charlotte's taxable income.

5. Given the points $A(-3, 2)$ and $B(5, 8)$.

- (a) Find the midpoint of interval AB .

1

- (b) Hence, find the equation of the perpendicular bisector of the interval AB .
Give your answer in the general form.

2

6. A bag contains 5 yellow marbles and 4 red marbles. Two draws are made without replacement. Calculate the probability of drawing:

(a) two yellow marbles 1

(b) at most one yellow marble. 1

7. Completely factorise the following

(a) $4x^2 - 9$ 1

(b) $x^2 - 5x - 24$ 1

(c) $6x^2 - 7x - 20$ 2

(d) $x^2 - y^2 + 4y - 4$ 2

8. A plane is rising at an angle of 15° inclined to the horizontal.
- (a) How far will the plane have to travel along its line of flight to increase its altitude (height) by 500 m, correct to 2 decimal places? 2
-
-
-
-
-
-
-
-
- (b) At what angle must the plane climb to achieve an increase in altitude of 500 m in half the distance needed at an angle of 15° , correct to the nearest minute? 2

Part D (24 marks)

1. Six years ago, Lydia was three times as old as Keiko. In three years time, Lydia will be twice as old as Keiko. What are their present ages? 2

2. Factorise completely $x^4 - 9x^3 - 8x + 72$ 2

3. Simplify $\frac{3^{2n-1} \times 27^{n+1}}{9^{3n}}$. 2

4. Simplify

(a) $\sqrt{300} - \sqrt{48} - \sqrt{147}$

1

(b) $a\sqrt{a^3} - \sqrt{4a^5}$

1

(c) $\left(\sqrt{a} - \frac{1}{\sqrt{a}}\right)^2 - \left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)^2$

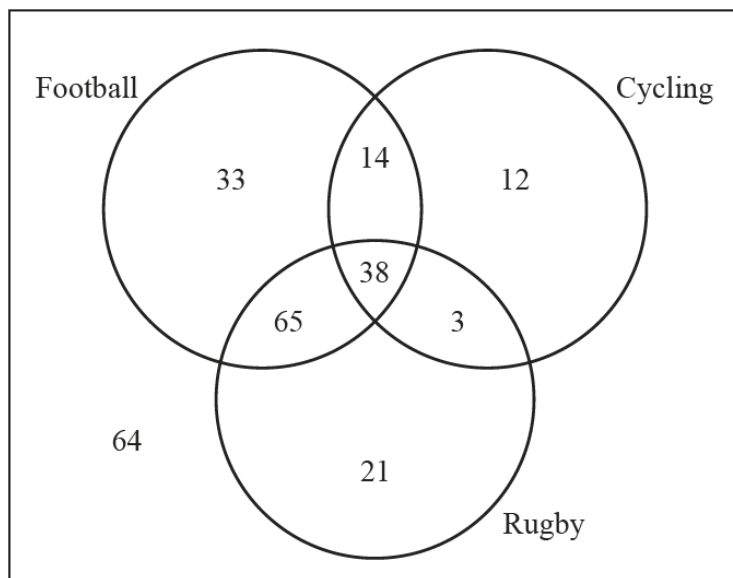
2

Part D continues on page 17

5. A survey is being carried out into the sports viewing habits of people in a particular area. As part of the survey, 250 people are asked which of the following sports they have watched on television in the past month.

- Football
- Cycling
- Rugby

The numbers of people who have watched these sports are shown in the Venn diagram.



One of the people is selected at random.

- (a) Find the probability that this person has in the past month watched cycling but not football. 1

- (b) Given that this person has watched cycling, find the probability that this person has not watched football 2

Part D continues on page 18

6. A food factory produces packets of candies which contain lollipops, sweets and chocolate bars in the ratios 5 : 6 : 7. In the following year, the number of lollipops is halved, the number of sweets is decreased by 10% and the number of chocolate is increased by 50%. Calculate the percentage increase or decrease in the total number of lollipops, sweets and chocolate bars in each packet.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slightly textured appearance and is set against a dark background.

7. Consider the following pattern.

$$S_1 = -2 + (1 \times 2) + (2 \times 2^2) = 8$$

$$S_2 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) = 32$$

$$S_3 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + (4 \times 2^4) = 96$$

$$S_4 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + (4 \times 2^4) + (5 \times 2^5) = 256$$

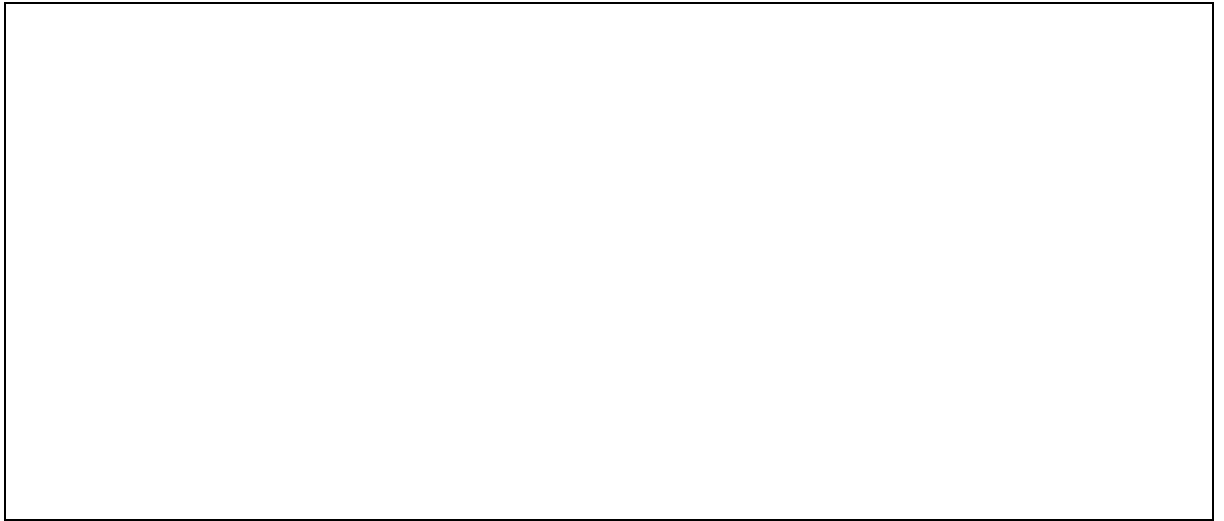
- (a) Write down the fifth line of the pattern. 1

- (b) By first considering the values of $\frac{1}{n}S_n$, write down the $(n-1)^{\text{th}}$ line of the pattern. 2

8. Two ships leave from the same port. One ship travels on a bearing of 157° at 20 knots. The second ship travels on a bearing of 247° at 35 knots. (1 knot is a speed of 1 nautical mile per hour.)

(a) Sketch the above information in a diagram below.

1



(b) How far apart are the ships after 8 hours, to the nearest nautical mile?

2

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(c) Calculate the bearing of the second ship from the first ship after 8 hours, to the nearest minute.

2

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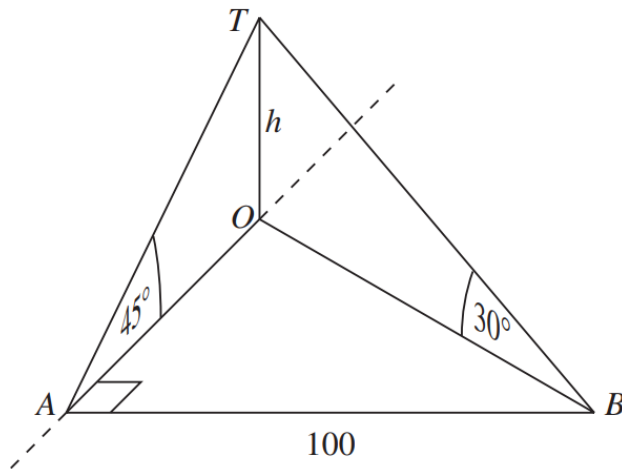
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End of Part D

Part E (22 marks)

1. A surveyor stands at a point A , which is due south of a tower OT of height h m. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100 m due east to point B , from where she measures the angle of elevation of the top of the tower to be 30° .



- (a) Express the length of OB in terms of h . 1

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- (b) Show that $h = 50\sqrt{2}$ m. 2

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- (c) Calculate the bearing of B from the base of the tower, correct to the nearest degree. 2

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2. Show that the points $L(-2, -13)$, $M(3, 2)$, $N(7, 14)$ are collinear.

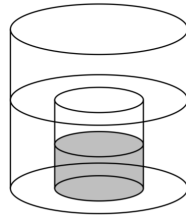
3

3. Find the co-ordinates of the point on the line $y = x - 9$ which is equidistant from the points $A(1, -4)$ and $B(5, -2)$.

3

Part E continues on page 22

4. Two vases are cylindrical in shape. The larger vase has diameter 20 cm. The smaller vase has diameter 10 cm and height 16 cm. The larger vase is partially filled with water. Then the empty smaller vase, with the open end at the top, is slowly pushed down into the water, which flows over its rim. When the smaller vase is pushed right down, it is half full of water. What was the original depth of the water in the larger vase? 3



5. A jar contains only red and yellow candies. Another 30 red candies are added to the candies already in the jar so that one-third of the total number of candies in the jar are red candies. At this point, 30 yellow candies are added to the jar, and now three-tenths of the total number of candies in the jar are red candies. What fraction of the number of the candies originally in the jar were red candies? 3

6. Candidates applying for jobs in a large company take an aptitude test, as a result of which they are either accepted, rejected, or retested, with probabilities 0.2, 0.5 and 0.3 respectively. When a candidate is retested for the first time, the three possible outcomes and their probabilities remain the same as for the original test.
When a candidate is retested for the second time there are just two possible outcomes, accepted or rejected, with probabilities 0.4 and 0.6 respectively.

(a) Draw a probability tree diagram to illustrate the outcomes.

1



(b) Find the probability that a randomly selected candidate is accepted.

2

(c) Find the probability that a randomly selected candidate is retested at least once, given that this candidate is accepted.

2

End of Exam



**SYDNEY
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2020

**YEAR 9
TERM 4
YEARLY ASSESSMENT TASK**

Mathematics Stage 5.3

Sample Solutions

NOTE:

**Some of you may be disappointed with your mark.
This process of checking your mark is NOT the opportunity
to improve your marks.
Improvement will come through further revision and practice,
as well as reading the solutions and comments.**

**Before putting in an appeal re marking, first consider that the mark is not
linked to the amount of ink you have used.**

**Just because you have shown 'working' does not justify that your
solution is worth any marks.**

Part A (22 marks)

For Questions 1 – 6, Circle the correct answer.

1. Stephen the author earns a percentage of the sales of his recent horror novel. Which word best describes this payment? 1

- (A) Salary
(B) Commission
(C) Royalty
(D) Wage

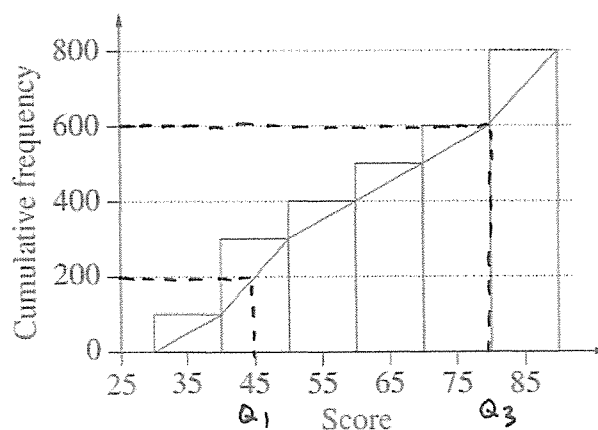
* Students commonly selected commission.
However, Stephen is not working at the bookstore selling each individual book.

2. The gradient and the y-intercept, respectively, of the line $2x + y - 3 = 0$ are? 1

- (A) 3 and -2
(B) -2 and 3
(C) 2 and -3
(D) -3 and -2

* Rearrange to $y = -2x + 3$
 $m = -2, c = 3$

3. A set of data is represented by the cumulative frequency histogram and ogive. 1



What is the best approximation for the interquartile range for this set of data?

- (A) 35
(B) 30
(C) 25
(D) 40

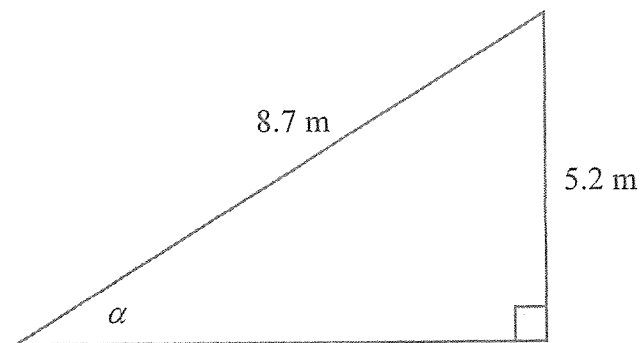
* Very few got this correct. Consider the annotated diagram, ignoring the histogram and focusing on the ogive.

4. Carlos took 70% of a whole pie. Maria took one third of the remainder. What percentage of the whole pie is left? 1

- (A) 30%
(B) 15%
(C) 10%
(D) 20%

* After Carlos took some, 30% is left.
Of this, Maria leaves two thirds.
 $\frac{2}{3} \times 30\% = 20\%$

5. Find the size of the angle α correct to the nearest minute. 1



- (A) $30^\circ 52'$
(B) $30^\circ 53'$
(C) $36^\circ 42'$
(D) $36^\circ 43'$

$$\begin{aligned} * \sin(\alpha) &= \frac{5.2}{8.7} \\ \alpha &= \sin^{-1}\left(\frac{5.2}{8.7}\right) \\ &= 36^\circ 42' 19.55'' \end{aligned}$$

6. Simplify $2m^2 \times 3mp^2$. 1

- (A) $5m^2p^2$
(B) $5m^3p^2$
(C) $6m^2p^2$
(D) $6m^3p^2$

$$* 2m^2 \times 3mp^2 = 2 \times m \times m \times 3 \times m \times p \times p$$

7. Express $9\,876\,000 \times 1\,234\,000$ in scientific notation correct to 3 significant figures. 2
 → in calculator: 1.2186984×10^{13} * Penalised if rounded incorrectly
 $= 1.22 \times 10^{13}$ (3 s.f.) and/or not expressed in scientific notation.

8. Simplify $9hk - j^2 + 5kh + 4j^2$. 1
 $= 9hk + 5hk + 4j^2 - j^2$ * Students penalised for not collecting
 $= 14hk + 3j^2$ all like terms and/or arithmetic errors

9. Expand and simplify $x - 3y - 4(y - 2x)$. 1
 $= x - 3y - 4y + 8x$ * Students penalised for errors
 $= 9x - 7y$ expanding and/or problems grouping terms.

10. Calculate $\frac{(3.2)^3}{6.8 + 4.7}$ correct to 4 decimal places. 1
 → in calculator $= 2.849391304$ * Students penalised for rounding
 $= 2.8494$ incorrectly.
 Many students used $(3.2)^2$ instead of $(3.2)^3$ for no mark.

11. A farmer spreads 25 kg of fertiliser over an area of 4000 m^2 . How much fertiliser would be needed to cover an area of 1.5 ha? 1
 $1.5 \text{ ha} = 15,000 \text{ m}^2$ * Many students did not convert hectares correctly.
 $\text{required fertiliser} = \frac{15,000}{4,000} \times 25$
 $= 93.75 \text{ kg}$

12. A car uses 18 L of petrol to travel 150 km. How much petrol would be needed to travel 350 km? 1

$$\text{required petrol} = \frac{350}{150} \times 18 \text{ L} \\ = 42 \text{ L}$$

* Well completed

13. Given that $a * b$ denotes $a^2 + ab - b^2$, find the value of $5 * (-2)$. 1

$$5 * (-2) = 5^2 + 5 \times (-2) - (-2)^2 \\ = 25 - 10 - 4 \\ = 11$$

* Well done, but students had difficulty manipulating $-(-2)^2$

14. Simplify

(a) $\frac{3x}{5} - \frac{x}{5} = \frac{2x}{5}$ 1

(b) $\frac{8a-4b}{4} = \frac{4(2a-b)}{4}$ 1

$$= 2a - b$$

(c) $\frac{2x-3}{5} - \frac{x-1}{2} = \frac{2(2x-3) - 5(x-1)}{5 \times 2}$ 2

$$= \frac{4x-6-5x+5}{10}$$
$$= \frac{-x-1}{10} \text{ or } -\frac{x+1}{10}$$

* Students had difficulty combining into 1 fraction and/or correctly expanding brackets.

(d) $\frac{1}{x} + \frac{2}{x+1}$

2

$$= \frac{1 \times (x+1)}{x(x+1)} + \frac{2 \times x}{x(x+1)}$$

$$= \frac{x+1+2x}{x(x+1)}$$

$$= \frac{3x+1}{x(x+1)} \text{ or } \frac{3x+1}{x^2+x}$$

* It is not necessary for (d) or (e) to expand the denominator, and led to mistakes and loss of marks for many students.

(e) $\frac{1}{2x+1} - \frac{1}{5x-1}$

2

$$= \frac{1 \times (5x-1) - 1 \times (2x+1)}{(2x+1)(5x-1)}$$

$$= \frac{5x-1-2x-1}{(2x+1)(5x-1)}$$

$$= \frac{3x-2}{(2x+1)(5x-1)} \text{ or } \frac{3x-2}{10x^2+3x-1}$$

End of Part A

2020 Yearly Exam Part B – Solutions and Comments

Solutions in pencil or with white out will **NOT** be remarked.

General feedback:

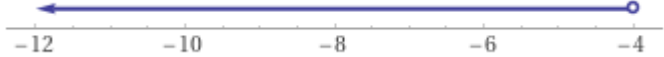
Disappointingly, no student scored full marks in Part B, with students not reading the questions carefully being a major factor contributing to this poor performance overall.

- Q1: A few students didn't draw a number line.
- Q4C: Many students didn't comment on the mode.
- Q6: Some students expanded, then factorised back to the original expression.
- Q7: Many students didn't calculate one of either the surface area or volume.

~

1. Solve $5 - 2x > 13$, and display your solution on a number line.

2

Solution	Comment(s)
$5 - 2x > 13$ $-2x > 8$ $x < -4$ 	Common error(s): • Not changing the direction of the inequality when dividing by -4. • Not using an open circle to exclude -4.

2. If $16m^2 + 24m + P = (4m + Q)^2$, what is the value for P ?

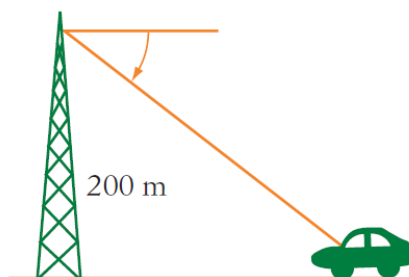
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Solution	Comment(s)
$(4m + Q)^2 = 16m^2 + 8Qm + Q^2$ Comparing coefficients of m: $8Q = 24$ $\therefore Q = 3$ $P = Q^2$ $= 9$	Common error(s): • Incorrectly expanding $(4m + Q)^2$. Most students didn't understand that this question involves comparing coefficients of terms. Students who assumed $Q = 3$ or found it through trial and error could only score a maximum of 1 mark.

2020 Yearly Exam Part B – Solutions and Comments

3. From the top of a 200 metre tower, the angle of depression of a car is 49° .
How far (to the nearest metre) is the car from the base of the tower?

1



Solution	Comment(s)
The angle of elevation is the same as the angle of depression, hence: $\therefore \tan 49^\circ = \frac{200}{x}$ $x = \frac{200}{\tan 49^\circ}$ $= 174 \text{ m (n.m.)}$	Alternate solution: Since the horizontal is perpendicular to the tower: $\tan 41^\circ = \frac{x}{200}$ $x = 200 \tan 41^\circ$ $= 174 \text{ m (n.m.)}$ No half marks awarded.

2020 Yearly Exam Part B – Solutions and Comments

Solutions in pencil or with white out will **NOT** be remarked.

4. At a tourist information office, the numbers of people seeking information each hour over the course of a 12-hour day are shown below.

6 25 38 39 31 18 35 31 33 15 21 28

- a. Construct a sorted stem and leaf diagram to represent these data.

1

Solution		Comment(s)
Stem	Leaf	No half marks awarded. Students who misplaced more than 2 scores could not score any marks.
0	6	
1	5 8	
2	1 5 8	
3	1 1 3 5 8 9	

- b. State the type of skewness suggested by your stem and leaf diagram.

1

Solution	Comment(s)
Negatively skewed.	No half marks awarded. Many students mistook a concentration of data to the right to mean positive skewness.

- c. For these data, find the median, mean and mode.

4

Comment on the usefulness of the mode in this case.

Solution	Comment(s)
The mode is the most common score: $Mode = 31$ The median is between the 6th and 7th score: $Median = \frac{28 + 31}{2}$ $= 29.5$ Using Statistics mode on the calculator, the mean is: $\bar{x} = 26.\dot{6}$ The mode has a frequency of 2, while all other scores have a frequency of 1. Hence, the mode isn't useful, as the difference in frequency isn't high enough for it to be meaningful.	Alternate solution for the mean using the formula: $\bar{x} = \frac{\sum fx}{\sum f}$$= \frac{320}{12}$$= 26.\dot{6}$ 1 mark per correct measure of central tendency. 1 mark for an appropriate comment on the mode. No half marks awarded. Most students either: • Forgot to comment on the mode, indicating that they didn't read the question. • Didn't know how to comment on the mode, indicating an inability to interpret data meaningfully.

2020 Yearly Exam Part B – Solutions and Comments

Solutions in pencil or with white out will **NOT** be remarked.

5. Simplify and write without negative indices.

a. $\left[-\left(\frac{1}{a}\right)^{-2} \right]^{-3}$

2

Solution	Comment(s)
$\left(-\left(\frac{1}{a}\right)^{-2} \right)^{-3} = -\left(\frac{1}{a}\right)^6$ $= -\frac{1}{a^6}$	Most students who rewrote the negative indices into fractions struggled to simplify correctly.

b. $\frac{(27a^2b^{-3})^{-1}}{(15ab^{-2})^{-2}}$

2

Solution	Comment(s)
$\frac{(27a^2b^{-3})^{-1}}{(15ab^{-2})^{-2}} = \frac{27^{-1}a^{-2}b^3}{15^{-2}a^{-2}b^4}$ $= \frac{25}{3b}$	Most students who rewrote the negative indices into fractions struggled to simplify correctly.

6. Expand and simplify

a. $(x + 5)(2 - 3x)$

1

Solution	Comment(s)
$(x + 5)(2 - 3x) = 2x - 3x^2 + 10 - 15x$ $= -3x^2 - 13x + 10$	Students generally performed well in this question.

b. $(2y + 3)^2$

1

Solution	Comment(s)
$(2y + 3)^2 = 4y^2 + 12y + 9$	Students generally performed well in this question.

c. $(2x - 3)(2x + 3)$

1

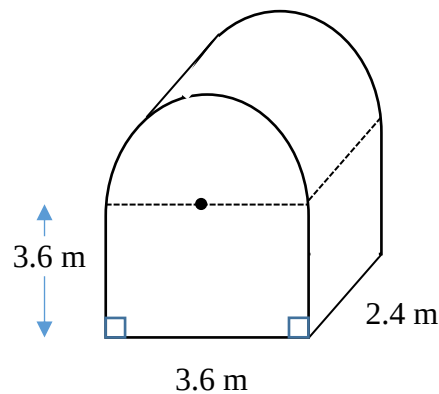
Solution	Comment(s)
$(2x - 3)(2x + 3) = 4x^2 - 9$	Students generally performed well in this question.

2020 Yearly Exam Part B – Solutions and Comments

Solutions in pencil or with white out will **NOT** be remarked.

7. Calculate the surface area and volume of this enclosed solid, correct to 1 decimal place.

4



Solution	Comment(s)
Volume: $V = Ah$ $= h \left(lb + \frac{\pi r^2}{2} \right)$ $= 2.4 \left(3.6^2 + \frac{\pi \times 1.8^2}{2} \right)$ $= 43.3 \text{ m}^3 \text{ (1 d.p.)}$ Surface area: $SA = (2lb + 2bh + lh) + \left(2 \times \frac{\pi r^2}{2} + \frac{2\pi r l}{2} \right)$ $= (2lb + 2bh + lh) + (\pi r^2 + \pi r l)$ $= (2 \times 3.6^2 + 2 \times 3.6 \times 2.4 + 3.6 \times 2.4)$ $+ (\pi \times 1.8^2 + \pi \times 1.8 \times 2.4)$ $= 75.6 \text{ m}^2 \text{ (1 d.p.)}$	1 mark for volume. 3 marks for surface area. Common error(s): Not halving the cylinder surface area and volume. Disappointingly, many students clearly didn't read the question, as they didn't calculate either the surface area or volume.

Part C (24 marks)

1. Solve $y = 4x + 3$ and $y = 17 - 3x$ simultaneously.

2

$$4x + 3 = 17 - 3x$$

$$7x + 3 = 17$$

$$7x = 14$$

$$x = 2 \quad (1)$$

Sub $x = 2$ into either equation

$$y = 11 \quad (1)$$

and stopped

Some students only worked out the x value. This will cost them 1 mark.

2 marks are awarded for correct working and correct answers.

2. If $y - \frac{1}{y} = 2\sqrt{6}$, then evaluate $y^2 + \frac{1}{y^2}$.

2

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$\begin{aligned} \left(y - \frac{1}{y}\right)^2 &= y^2 - 2 \times y \times \frac{1}{y} + \frac{1}{y^2} \\ &= y^2 - 2 + \frac{1}{y^2} \end{aligned}$$

$$\left(y - \frac{1}{y}\right)^2 = y^2 - 2 + \frac{1}{y^2}$$

$$\left(y - \frac{1}{y}\right)^2 + 2 = y^2 - 2 + \frac{1}{y^2} + 2$$

$$\left(y - \frac{1}{y}\right)^2 + 2 = y^2 + \frac{1}{y^2}$$

Only 1 mark for $(2\sqrt{6})^2 = 24$ or showing understanding of perfect square but incorrect answer

$$\begin{aligned} y^2 + \frac{1}{y^2} &= (2\sqrt{6})^2 + 2 \\ &= 26 \quad (2) \end{aligned}$$

3. If $y = 5^x$. Express 5^{2x+1} in terms of y .

2

$$5^{2x+1} = 5^x \times 5^x \times 5$$

$$\begin{aligned} 5^{2x+1} &= (5^x)^2 \times 5 \\ &= 5y^2 \end{aligned}$$

$$\begin{aligned} &= 5^x \times 5^x \times 5 \quad \text{or} \\ &= y^2 \times 5 \\ &= 5y^2 \quad (2) \end{aligned}$$

1 mark only for y^2 or an expression containing y^2 .

4. The table shows the income tax rates for the 2019–2020 financial year.

3

Taxable income	Tax payable on this income
\$0 to \$18 200	Nil
\$18 201 to \$37 000	19c for each \$1 over \$18 200
\$37 001 to \$90 000	\$3 572 plus 32.5c for each \$1 over \$37 000
\$90 001 to \$180 000	\$20 797 plus 37c for each \$1 over \$90 000
\$180 001 and over	\$54 097 plus 45c for each \$1 over \$180 000

(Note: The above rates do not include the Medicare levy of 2%.)

The Medicare levy is calculated as 2% of taxable income.

For the 2019–2020 financial year, Charlotte pays a Medicare levy of \$2341.20.

Calculate the tax payable on Charlotte's taxable income.

$$2341.20 \div 2\% = \$117060$$

$$\text{Taxable Income} = \$117060$$

$$\text{Tax} = 20797 + 0.37(117060 - 90000) \\ = \$30809.2$$

1 mark for correct taxable income.

1 mark for correct tax bracket AND correct operation

1 mark for correct final answer.

5. Given the points $A(-3, 2)$ and $B(5, 8)$.

- (a) Find the midpoint of interval AB .

1

$$\frac{-3+5}{2} = 1$$

$\frac{1}{2}$ mark for each correct value

$$\frac{2+8}{2} = 5$$

$(1, 5)$

- (b) Hence, find the equation of the perpendicular bisector of the interval AB .
Give your answer in the general form.

2

$$m_{AB} = \frac{8-2}{5-(-3)}$$

$$= \frac{3}{4}$$

$$m_{\text{bisector}} = -\frac{4}{3}$$

$$y-5 = -\frac{4}{3}(x-1)$$

$$3y-15 = -4x+4$$

$$4x+3y-19=0 \quad (1)$$

$\left(\frac{1}{2}\right)$

if equation correct
but not in general form
ie $y = -\frac{4}{3}x + \frac{19}{3}$

$\left(\frac{1}{2}\right)$

6. A bag contains 5 yellow marbles and 4 red marbles. Two draws are made without replacement. Calculate the probability of drawing:

(a) two yellow marbles

1

$$\frac{5}{9} \times \frac{4}{8} = \frac{5}{18} \quad (1)$$

(b) at most one yellow marble.

1

$$1 - P(2 \text{ yellows})$$

$$= \frac{13}{18} \quad (1)$$

7. Completely factorise the following

(a) $4x^2 - 9$

1

$$(2x+3)(2x-3) \quad (1)$$

must be entirely correct
for the 1 mark

(b) $x^2 - 5x - 24$

1

$$(x-8)(x+3) \quad (1)$$

(c) $6x^2 - 7x - 20$

2

$$\begin{array}{cc} 3x & 4 \\ \times & \\ 2x & -5 \end{array}$$

$$-5 \times 3x + 4 \times 2x = -7x$$

$$(3x+4)(2x-5)$$

Some wrote $(3x-5)(2x+4)$, 1 mark
awarded

(d) $x^2 - y^2 + 4y - 4$

2

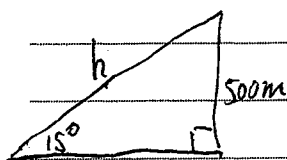
$$= x^2 - (y^2 - 4y + 4)$$

$$= x^2 - (y-2)^2 \quad (1)$$

$$= (x+y-2)(x-y+2) \quad (1)$$

8. A plane is rising at an angle of 15° inclined to the horizontal.

- (a) How far will the plane have to travel along its line of flight to increase its altitude (height) by 500 m, correct to 2 decimal places? 2



$$\sin 15^\circ = \frac{500}{h} \quad (1)$$

$$h = 500 \div \sin 15^\circ$$

$$= 1931.85 \text{ m} \quad (1)$$

no marks awarded if trig ratio was incorrect, ie, tan or cos

- (b) At what angle must the plane climb to achieve an increase in altitude of 500 m in half the distance needed at an angle of 15° , correct to the nearest minute? 2
flight path

$$\sin \theta = \frac{500}{0.5h} \quad (1)$$

$$\theta = \sin^{-1} \left(\frac{500}{0.5h} \right)$$

$$= 31^\circ 10' \quad (1)$$

If incorrect trig ratio was already penalised in part (a), no more penalty for the same mistake

Part D (24 marks)

1. Six years ago, Lydia was three times as old as Keiko. In three years time, Lydia will be twice as old as Keiko. What are their present ages? 2

$$\begin{aligned} \textcircled{1} L - 6 &= 3(K - 6) \quad \checkmark \frac{1}{2} \\ L &= 3K - 12 \\ \textcircled{2} L + 3 &= 2(K + 3) \quad \checkmark \frac{1}{2} \\ &= 2K + 6 \\ \textcircled{1} &= \textcircled{2} \quad 3K - 12 = 2K + 6 \\ K &= 15, \quad L = 30 + 3 = 33 \quad \checkmark \frac{1}{2} \end{aligned}$$

2. Factorise completely $x^4 - 9x^3 - 8x + 72$ 2

$$\begin{aligned} &= x^3(x - 9) - 8(x - 9) \quad \checkmark \frac{1}{2} \\ &= (x - 9)(x^3 - 8) \quad \checkmark \frac{1}{2} \\ &= (x - 9)(x - 2)(x^2 + 2x + 4) \quad \checkmark 1 \end{aligned}$$

3. Simplify $\frac{3^{2n-1} \times 27^{n+1}}{9^{3n}}$ 2

$$\begin{aligned} &= \frac{3^{2n-1} \times 3^{3(n+1)}}{3^{2(3n)}} \\ &= \frac{3^{5n+2}}{3^{6n}} \quad \checkmark \frac{1}{2} \\ &= 3^{-n+2} \quad \checkmark 1 \\ &= \frac{9}{3^n} \quad \text{for either} \end{aligned}$$

4. Simplify

(a) $\sqrt{300} - \sqrt{48} - \sqrt{147}$

$$= 10\sqrt{3} - 4\sqrt{3} - 7\sqrt{3} = -\sqrt{3}$$

1

(b) $a\sqrt{a^3} - \sqrt{4a^5}$

$$\begin{aligned} &= a^1 \times a^{3/2} - 2a^{5/2} \\ &= a^{5/2} - 2a^{5/2} \\ &= -a^{5/2} \\ \text{OR } &= -\sqrt{a^5} \end{aligned}$$

OR Radical Form

$$\begin{aligned} &= a^2 \sqrt{a} - 2a^2 \sqrt{a} \\ &= -a^2 \sqrt{a} \\ &= -\sqrt{a^4} \sqrt{a} \\ &= -\sqrt{a^5} \end{aligned}$$

1

(c) $\left(\sqrt{a} - \frac{1}{\sqrt{a}}\right)^2 - \left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)^2$

$$\begin{aligned} &= a - \frac{2\sqrt{a}}{\sqrt{a}} + \frac{1}{a} - \left(a + \frac{2\sqrt{a}}{\sqrt{a}} + \frac{1}{a}\right) \\ &= -2 - 2 = -4 \end{aligned}$$

OR difference of two squares:

$$\left[\left(\sqrt{a} - \frac{1}{\sqrt{a}}\right) + \left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)\right] \cdot \left[\left(\sqrt{a} - \frac{1}{\sqrt{a}}\right) - \left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)\right]$$

$$= 2(\sqrt{a})^2 \times -\frac{2}{(\sqrt{a})^2}$$

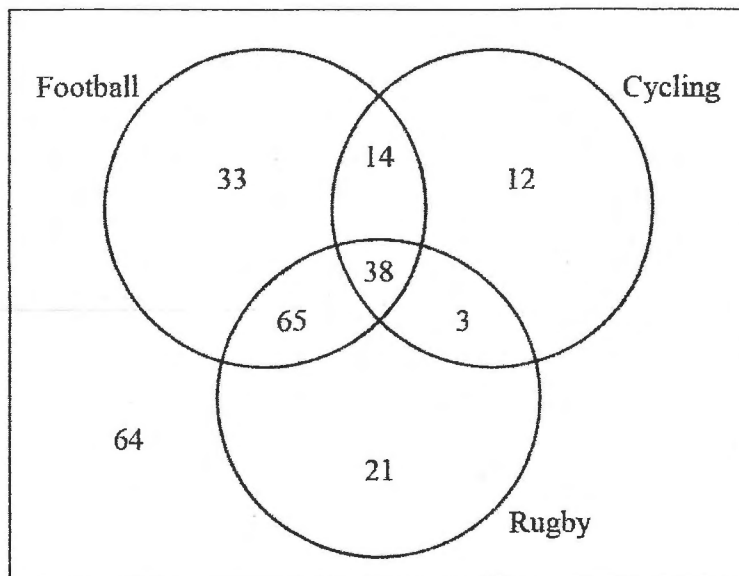
$$= -4$$

Part D continues on page 17

5. A survey is being carried out into the sports viewing habits of people in a particular area. As part of the survey, 250 people are asked which of the following sports they have watched on television in the past month.

- Football
- Cycling
- Rugby

The numbers of people who have watched these sports are shown in the Venn diagram.



One of the people is selected at random.

- (a) Find the probability that this person has in the past month watched cycling but not football. 1

$$= \frac{12+3}{250} = \frac{15}{250} = \frac{3}{50} \checkmark$$

* full marks given if not simplified.

- (b) Given that this person has watched cycling, find the probability that this person has not watched football 2

$S = \text{People who watch cycling}$

$$= \frac{3+12}{14+38+3+12}$$

$$= \frac{15}{67} \checkmark$$

Part D continues on page 18

6. A food factory produces packets of candies which contain lollipops, sweets and chocolate bars in the ratios 5 : 6 : 7. In the following year, the number of lollipops is halved, the number of sweets is decreased by 10% and the number of chocolate is increased by 50%. Calculate the percentage increase or decrease in the total number of lollipops, sweets and chocolate bars in each packet.

3

$$L:S:C = 5:6:7$$

$$\text{Following Year: } 2.5:5.4:10.5 \checkmark$$

$$\text{Total} = 18 \checkmark \frac{1}{2}$$

$$\text{Total (following year)} = 18.4 \checkmark \frac{1}{2}$$

$$\% \Delta = \frac{0.4}{18} \times 100 = 2.2\% \text{ (1 dp)} \checkmark$$

Notes: - $\frac{1}{2}$ deducted for answers left in decimal or fraction form

7. Consider the following pattern.

$$S_1 = -2 + (1 \times 2) + (2 \times 2^2) = 8$$

$$S_2 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) = 32$$

$$S_3 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + (4 \times 2^4) = 96$$

$$S_4 = -2 + (1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + (4 \times 2^4) + (5 \times 2^5) = 256$$

- (a) Write down the fifth line of the pattern.

$$S_5 = -2 + (1 \times 2) + \dots + (6 \times 2^6) = 640 \checkmark \frac{1}{2}^1$$

- (b) By first considering the values of $\frac{1}{n} S_n$, write down the $(n-1)^{\text{th}}$ line of the pattern.

1 mark for at least two of these

$$\frac{1}{1} S_1 = 8$$

$$\frac{1}{2} S_2 = 16$$

$$\frac{1}{3} S_3 = 32$$

$$\frac{1}{4} S_4 = 64$$

$$\therefore \frac{1}{n} S_n = 2^{n+2} \checkmark \frac{1}{2}$$

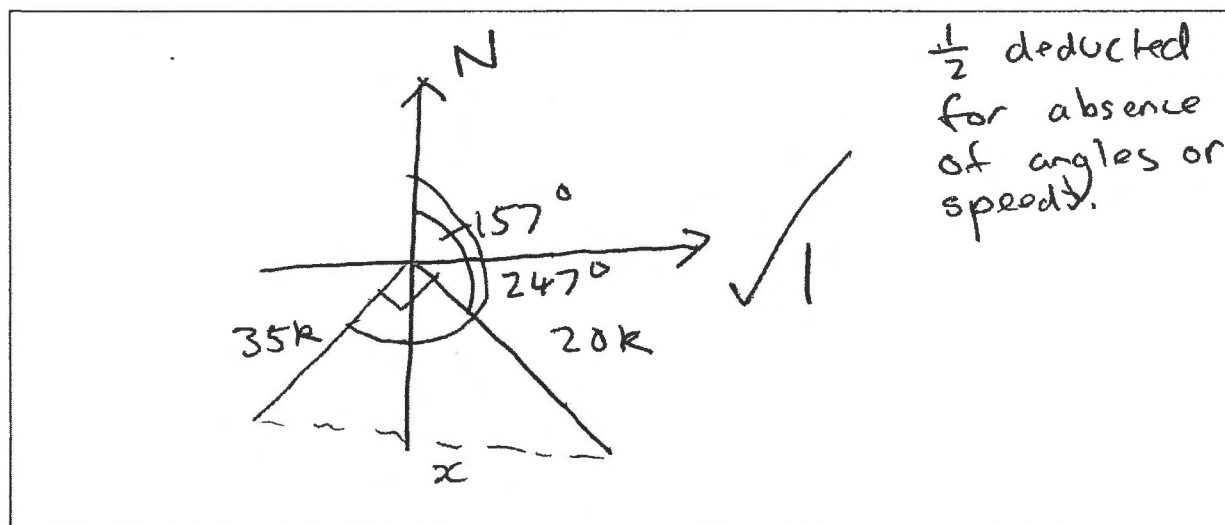
$$\rightarrow S_n = n \times 2^{n+2}$$

$$\therefore S_{n-1} = (n-1) 2^{n+1} \checkmark \frac{1}{2}$$

8. Two ships leave from the same port. One ship travels on a bearing of 157° at 20 knots. The second ship travels on a bearing of 247° at 35 knots. (1 knot is a speed of 1 nautical mile per hour.)

(a) Sketch the above information in a diagram below.

1



(b) How far apart are the ships after 8 hours, to the nearest nautical mile?

2

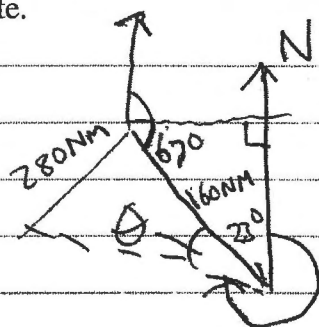
$$d_1 = 20 \times 8 = 160, \quad d_2 = 35 \times 8 = 280$$

$$x = \sqrt{160^2 + 280^2} = 322 \text{ NM}$$

no marks deducted for answer in surd/exact form.

(c) Calculate the bearing of the second ship from the first ship after 8 hours, to the nearest minute.

2



$$\tan \theta = \frac{280}{160}$$

$$\theta = \tan^{-1}\left(\frac{280}{160}\right)$$

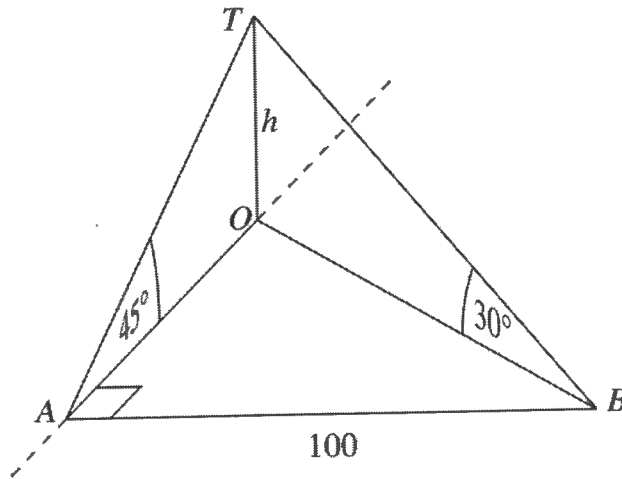
$$= 60^\circ 15'$$

$$\text{Bearing: } 360 - (23 + 60^\circ 15') = 276^\circ 45'$$

End of Part D

Part E (22 marks)

1. A surveyor stands at a point A , which is due south of a tower OT of height h m. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100 m due east to point B , from where she measures the angle of elevation of the top of the tower to be 30° .



- (a) Express the length of OB in terms of h .

1

In $\triangle OBT$ $\tan 30 = \frac{h}{OB}$

$$\Rightarrow OB = \frac{h}{\tan 30} = \frac{h}{\frac{1}{\sqrt{3}}}$$

$$OB = \sqrt{3}h$$

- (b) Show that $h = 50\sqrt{2}$ m.

2

In $\triangle AOT$, $\tan 45 = \frac{h}{OA}$

$$\Rightarrow 1 = \frac{h}{OA}$$

$$\Rightarrow OA = h$$

Then $OA^2 + AB^2 = OB^2$

$$h^2 + 100^2 = 3h^2$$

$$2h^2 = 100^2$$

$$h^2 = 5000$$

$$h = 50\sqrt{2}$$

- (c) Calculate the bearing of B from the base of the tower, correct to the nearest degree.

2

$$\tan \angle AOB = \frac{100}{h} = \frac{100}{50\sqrt{2}} = \sqrt{2}$$

$$\therefore \angle AOB = 54.7^\circ$$

$$= 55^\circ$$

Then bearing from tower to $B = \angle NOB$

$$= 180^\circ - 55^\circ$$

$$= 125^\circ \text{ N}$$

2. Show that the points $L(-2, -13)$, $M(3, 2)$, $N(7, 14)$ are collinear.

3

$$m_{LM} = \frac{2 - (-13)}{3 - (-2)} = 3$$

$$m_{MN} = \frac{14 - 2}{7 - 3} = 3$$

Since gradients are equal and M is common.
then L, M, N are collinear.

3

3. Find the co-ordinates of the point on the line $y = x - 9$ which is equidistant from the points $A(1, -4)$ and $B(5, -2)$.

3

Let C lie on $y = x - 9 \Rightarrow$ Coordinates of $C = (x, x - 9)$

The $d_{AC}^2 = d_{BC}^2$ Distance

$$\Rightarrow (x - 1)^2 + (x - 9 - (-4))^2 = (x - 5)^2 + (x - 9 - (-2))^2$$

$$(x - 1)^2 + (x - 5)^2 = (x - 5)^2 + (x - 7)^2$$

$$\Rightarrow (x - 1)^2 = (x - 7)^2$$

$$x^2 - 2x + 1 = x^2 - 14x + 49$$

$$12x = 48$$

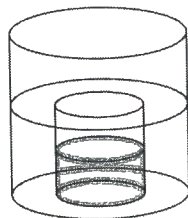
$$x = 4$$

Then co-ordinates of $C = (4, -5)$

3

Part E continues on page 22

4. Two vases are cylindrical in shape. The larger vase has diameter 20 cm. The smaller vase has diameter 10 cm and height 16 cm. The larger vase is partially filled with water. Then the empty smaller vase, with the open end at the top, is slowly pushed down into the water, which flows over its rim. When the smaller vase is pushed right down, it is half full of water. What was the original depth of the water in the larger vase? 3



$$V_{\text{small cylinder}} = \pi r^2 h = \pi 25 \times 16 = 400\pi \text{ cm}^3$$

$$\text{This fills to half} \Rightarrow V = 200\pi \text{ cm}^3$$

At this point water in big cylinder reaches top of small cylinder

$$\therefore \text{Original volume of water in big cylinder} = \text{Vol big cylinder at } h=16\text{cm} \text{ plus } 200\pi$$

$$\text{Orig. Vol. big cyl.} = \pi 100 \times 16 + 200\pi = 1800\pi \text{ cm}^3 \text{ (of water)}$$

$$\text{Then } h = \frac{V}{\pi r^2} = \frac{1800\pi}{\pi \times 100} = \underline{18\text{cm}}$$

5. A jar contains only red and yellow candies. Another 30 red candies are added to the candies already in the jar so that one-third of the total number of candies in the jar are red candies. At this point, 30 yellow candies are added to the jar, and now three-tenths of the total number of candies in the jar are red candies. What fraction of the number of the candies originally in the jar were red candies? 3

Let r = no. red, y = no. yellow

$$\text{Add 30 red} \Rightarrow \frac{\text{Red}}{\text{Total}} = \frac{1}{3} = \frac{r+30}{r+y+30}$$

$$\Rightarrow r+30 = \frac{r+y+30}{3} \quad (1)$$

$$\text{Add 30 yellow} \Rightarrow \frac{\text{Red}}{\text{Total}} = \frac{3}{10} = \frac{r+30}{r+y+60} \quad (2)$$

$$\text{Sub (1) in (2)} \Rightarrow \frac{r+y+30}{3(r+y+60)} = \frac{3}{10}$$

$$\Rightarrow 10(r+y+30) = 9(r+y+60)$$

$$10r + 10y + 300 = 9r + 9y + 540$$

$$r+y = 240 \quad (3)$$

$$\text{Sub in (1)} \Rightarrow \frac{r+30}{240+30} = \frac{1}{3}$$

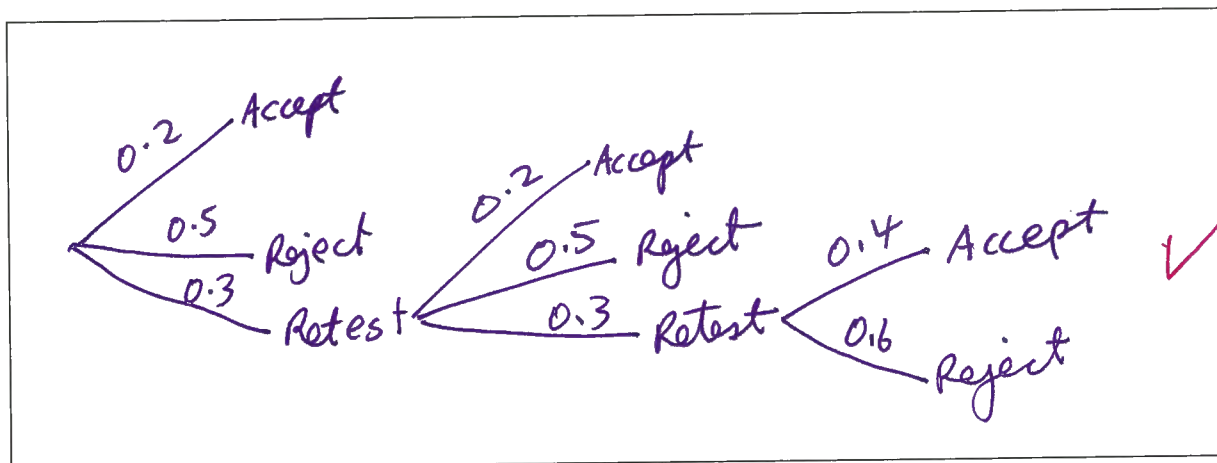
$$\begin{aligned} \text{Sub in (3)} &\Rightarrow y = 180 \\ \text{Then initial fraction of red} &= \frac{60}{240} \\ &= \underline{\underline{\frac{1}{4}}} \end{aligned}$$

Candidates applying for jobs in a large company take an aptitude test, as a result of which they are either accepted, rejected, or retested, with probabilities 0.2, 0.5 and 0.3 respectively. When a candidate is retested for the first time, the three possible outcomes and their probabilities remain the same as for the original test.

When a candidate is retested for the second time there are just two possible outcomes, accepted or rejected, with probabilities 0.4 and 0.6 respectively.

- (a) Draw a probability tree diagram to illustrate the outcomes.

1



- (b) Find the probability that a randomly selected candidate is accepted.

2

$$\begin{aligned}
 P(\text{Accepted}) &= P(\text{Accepted or Retested and Accepted or Retested, Retested, Accepted}) \\
 &= 0.2 + 0.3 \times 0.2 + 0.3 \times 0.3 \times 0.4 \\
 &= 0.2 + 0.06 + 0.036 \\
 &= 0.296 \quad \left(\text{or } \frac{37}{125}\right)
 \end{aligned}$$

- (c) Find the probability that a randomly selected candidate is retested at least once, given that this candidate is accepted.

2

$$\begin{aligned}
 &P(\text{retested once or twice given they were accepted}) \\
 &= \frac{P(\text{retested and accepted})}{P(\text{accepted})} \\
 &= \frac{0.3 \times 0.2 + 0.3^2 \times 0.4}{0.296} = \frac{0.096}{0.296} = \frac{12}{37}
 \end{aligned}$$

End of Exam